



M002 Week 5 — Mini Cube Puzzle

Topic: Mini Cube Puzzle · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

Finish the mini cube. Add a conditional, fill every layer, then break the top and climb out.

1 · Predict 🧠

Read the steps. Write what happens.

```
repeat 5 times:
  place block
  move forward
```

What shape is left? How long?


```
if on the edge OR on the top layer:
  place block
otherwise:
  leave empty
```

Only one part must be true. Which spots get a block?

2 · Plain to Code

The agent should move up one block. Match the plain words to the API.

```
move up one block
```

Word bank: `agent.move(UP, 1)` · `agent.move(FORWARD, 1)` · `agent.turn(LEFT_TURN)`

Write the line:

3 · Trace 🐾

```
agent.place(BLOCK)
agent.move(FORWARD, 1)
agent.place(BLOCK)
agent.move(UP, 1)
```

How many blocks placed? Where does the agent end — on the floor or one up?

4 · Spot the Bug 🐛

Bug A — The agent should fill the layer, **then** move up. It moves up too soon.

```
agent.place(BLOCK)
agent.move(UP, 1)
agent.place(BLOCK)
```

Fix it:

Bug B — To climb out, make the hole **first**, then move up. Fix the order.

```
agent.move(UP, 1)
agent.detect(BLOCK, UP)
```

Hint: break the block above first.

Fix it:

5 · Finish the Cube 📷 🎥

Open this week's world. Find the **mini cube starter** — part is built.

1. Keep the starter going.
2. Add one conditional that uses **OR**.
3. Fill every layer.
4. Break a hole in the top and climb out.

Send a photo or video of the finished cube and the hole. Then finish:

To keep building, I ...

I used a conditional when ...

I made the hole by ...

Phone video plan. Film the agent building. Use these words: **cube, layer, conditional, OR, hole.**

1. Show the starter cube.
2. Run your code and watch it fill.
3. Read your conditional out loud.
4. Show the hole and the climb out.

Write what you will say:

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.